

SATTAHIP, Thailand

WMO: 484770

Lat: 12.6932N

Lon: 100.9820E

Elev: 70

StdP: 14.66

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	66.8	69.2	53.0	60.0	75.6	56.4	68.0	76.7	13.4	80.5	12.1	82.4	4.8	0	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.2	94.3	80.3	93.2	80.4	92.3	80.1	83.6	90.3	82.9	89.5	82.4	88.9	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.1	167.3	88.0	81.0	161.7	87.2	80.7	159.9	86.9	47.8	90.7	47.0	89.9	46.4	88.9	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.5	12.1	11.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			62.6	98.7	3.2	2.5	60.4	100.5	58.5	102.0	56.7	103.5	54.4	105.3
			56.9	85.6	3.2	1.8	54.7	86.9	52.8	87.9	51.0	88.9	48.7	90.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.7	80.2	82.1	84.2	86.3	86.2	85.5	85.0	84.8	83.7	82.9	82.5	80.7	(6)
(7)		DBStd	3.00	2.66	2.47	2.46	2.38	2.31	1.95	2.04	1.76	1.94	1.70	2.07	3.08	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12291	935	898	1059	1090	1122	1066	1083	1080	1012	1021	975	950	(10)
(11)		CDD65	6816	470	478	594	640	657	616	618	615	562	556	525	485	(11)
(12)		CDH74	81147	4654	5338	7365	8623	8812	8081	7832	7663	6602	5921	5481	4775	(12)
(13)		CDH80	35517	1771	2101	3248	4436	4441	3861	3494	3324	2585	2151	2185	1920	(13)
(14)	Wind	WSAvg	5.4	4.8	5.5	6.3	6.0	5.6	6.0	6.1	6.0	4.7	4.1	4.8	5.3	(14)
(15)	Precipitation	PrecAvg	49.0	1.0	0.9	2.3	3.0	5.9	5.9	4.6	4.1	8.8	9.6	2.4	0.4	(15)
(16)		PrecMax	64.8	4.6	3.5	8.4	9.3	12.5	14.8	10.5	10.4	15.2	16.2	7.6	1.7	(16)
(17)		PrecMin	36.3	0.0	0.0	0.0	0.1	1.2	0.5	0.8	1.8	3.6	4.1	0.0	0.0	(17)
(18)		PrecStd	7.3	1.2	0.9	1.9	2.2	3.1	3.3	2.7	2.1	3.0	3.1	2.1	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.3	92.4	93.6	96.1	95.6	94.8	93.5	93.3	93.0	92.9	93.5	93.5	(19)
(20)		MCWB	74.9	76.4	79.2	81.0	81.6	80.8	80.2	79.7	79.9	79.3	77.5	76.8	76.8	(20)
(21)		2%	DB	90.3	90.7	91.8	94.0	93.6	92.9	91.9	91.7	91.2	91.2	91.7	91.6	(21)
(22)		MCWB	75.4	78.0	80.1	81.4	81.7	80.7	80.0	79.7	79.7	79.4	77.7	76.5	76.5	(22)
(23)		5%	DB	89.2	89.6	90.9	92.8	92.6	91.5	90.7	90.6	89.8	89.9	90.6	89.9	(23)
(24)		MCWB	75.7	78.1	80.1	81.6	81.5	80.4	79.8	79.5	79.5	79.2	77.5	76.0	76.0	(24)
(25)		10%	DB	87.8	88.2	89.7	91.5	91.3	90.1	89.5	89.4	88.6	88.6	89.4	88.5	(25)
(26)		MCWB	75.5	77.6	79.8	81.2	80.8	80.1	79.6	79.4	79.3	79.0	77.2	75.3	75.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	82.0	83.4	84.6	84.9	83.6	83.5	83.1	82.8	82.7	82.5	81.1	(27)
(28)		MADB	87.4	88.3	89.5	91.7	92.2	89.7	88.8	88.3	88.2	88.7	88.9	88.3	88.3	(28)
(29)		2%	WB	79.9	80.9	82.1	83.5	83.6	82.7	82.4	81.9	81.8	81.7	81.1	79.8	(29)
(30)		MADB	86.3	87.3	89.0	90.6	90.8	88.9	87.9	87.4	87.3	87.7	88.0	87.3	87.3	(30)
(31)		5%	WB	79.0	80.1	81.3	82.7	82.7	82.0	81.6	81.1	81.0	80.9	80.0	78.7	(31)
(32)		MADB	85.3	86.4	88.2	90.1	89.6	88.2	87.4	86.9	86.7	86.9	86.8	86.3	86.3	(32)
(33)		10%	WB	78.1	79.4	80.6	82.0	82.0	81.4	80.9	80.4	80.3	80.1	79.0	77.5	(33)
(34)		MADB	84.2	85.7	87.5	89.3	88.7	87.7	86.9	86.5	86.0	85.7	85.4	84.8	84.8	(34)
(35)	Mean Daily Temperature Range	MDBR	16.5	13.9	11.6	11.2	10.9	10.5	9.8	10.2	10.8	12.5	14.8	16.8	16.8	(35)
(36)		5% DB	MADB	17.9	15.3	13.0	12.3	12.4	11.8	11.1	11.5	12.1	14.3	16.5	18.0	(36)
(37)		MCWBR	8.1	6.4	5.2	5.0	5.2	4.9	4.7	4.8	5.1	6.3	6.9	7.7	7.7	(37)
(38)		5% WB	MADB	14.6	13.1	11.8	11.5	11.7	11.0	10.2	10.9	11.5	13.0	14.5	16.1	(38)
(39)	MCWBR	7.4	6.0	5.4	5.3	5.5	5.2	5.0	5.2	5.7	6.3	7.1	7.7	7.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.535	0.585	0.592	0.536	0.474	0.456	0.454	0.454	0.463	0.531	0.496	0.484	(40)	
(41)		taud	2.016	1.902	1.929	2.107	2.332	2.399	2.399	2.386	2.344	2.114	2.196	2.195	(41)	
(42)		Ebn at Noon	244	237	239	251	263	265	266	269	269	248	254	256	(42)	
(43)		Edh at Noon	55	63	63	52	41	38	38	39	41	50	45	45	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1592	1735	1772	1860	1669	1595	1546	1582	1531	1507	1563	1560	(44)	
(45)		RadStd	104	89	136	122	113	85	91	106	90	104	121	82	82	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	+130	+130

Nomenclature: See separate page